AYUSH KUMAR

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Portfolio ◇ GitHub ◇ GitLab ◇ LinkedIn

EDUCATION

R V Institute of Technology and Management *Expected May 2027*
Bachelor of Engineering, Computer Science & Engineering CGPA: 8.99 / 10.00, through 6th Semester
Doon Public School, Dhanbad, Jharkhand *2022*
Class XII, CBSE Percentage: 78.6%

EXPERIENCE & OPEN SOURCE

Open Source Software Contributor *2023 – Present*
166 public contributions, 120 merged: 153 GitHub PRs and 13 GitLab MRs across **pyenv**, **SugarLabs**, **Boeing**, and **CDLI Framework**.

- Designed and tested fixes in Python and JavaScript for reproduced defects and CI failures across CLI workflows, APIs, and developer tooling.
- Collaborated with maintainers through issue discussions and review feedback, using test suites and CI results to keep fixes minimal, reviewable, and clearly communicated.

PROJECTS

TALON - Network Intrusion Detection and Traffic Analysis Platform *May 2026 – Jul 2026*
C++17, FastAPI, Python, XDP/eBPF, PostgreSQL, Redis Live Demo | GitHub

- Developed a C++17/libpcap packet-to-flow pipeline with bounded producer-consumer queues and five-tuple aggregation, producing ML-ready JSONL telemetry with TCP flag features.
- Analyzed and evaluated 10,000 capture-day-separated CICIDS2017 flows, reaching **86.09% accuracy**, **98.94% recall**, **78.71% precision**, and **0.8767 F1**.
- Benchmarked native XDP against the C++ parser across seven runs: **16.05M packets/s**, **1.77x throughput**, and **42.6% lower per-packet processing cost**.

SENTINEL - Distributed Market Microstructure Simulator *Feb 2026 – Aug 2026*
JavaScript, FastAPI, Python, XGBoost, scikit-learn Live Demo | GitHub

- Engineered a market simulator for load testing with **40+ concurrent agents**, live order-book analytics, telemetry, and risk/P&L dashboards.
- Built **8+ trader types** with latency injection and synchronization; used event traces to resolve race conditions and state drift during stress runs.
- Developed liquidity-shock and trading-pattern models reaching **85% accuracy** and **83% precision**; presented at **IEEE ICWITE 2026**.

ENGRAM - Self-Hostable AI Memory Layer for Cross-LLM Context Sync *May 2026*
Python, FastAPI, JavaScript, PostgreSQL, MCP Live Demo | GitHub

- Architected memory middleware across **4 LLM adapters** with extraction, deduplication, semantic search, confidence scoring, and access tracking behind a FastAPI proxy.
- Built a **7-tool JavaScript MCP server** and dashboard for memory CRUD, retrieval logs, and integration testing.
- Added asynchronous memory reconciliation and retrieval logging to make cross-adapter context updates observable and diagnosable.

PRISM - Parkinson's Disease Assessment Portal *2025*
Python, PyTorch, scikit-learn, XGBoost, Transformers, RAG GitHub

- Developed a Parkinson's disease assessment portal using PPMI patient data, symptom features, traditional ML models, and medical transformer models.
- Built prediction pipelines with LightGBM (**86.7% accuracy**), XGBoost, SVM, PubMedBERT, BioGPT, and Clinical-T5, combining outputs through a multimodal ensemble.
- Implemented RAG-based medical report generation using indexed literature, patient feature analysis, and PDF export.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, C
Systems & Backend: Linux, FastAPI, REST APIs, TCP/IP
Data & Systems: PostgreSQL, Redis, MongoDB, MySQL, Docker
AI / ML: PyTorch, XGBoost, scikit-learn, Hugging Face Transformers, MCP, NumPy, Pandas
Cloud & Tooling: AWS, Azure, Git, GitHub, GitLab, CI/CD
Core CS: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, Distributed Systems, OOP

ACHIEVEMENTS & LEADERSHIP

- **Events Head, Robofiesta 2025, National-Level Techfest:** led **24+ events**, **70+ team members**, and **500+ participants**.
- **Technical Head, E-Summit 2025**; led Robotics, Web Dev, Cybersecurity, Coding, and Entrepreneurship clubs.
- Presented SENTINEL at **IEEE ICWITE 2026**; Zenodo papers: soybean disease/yield and transformer tweet sentiment.